

NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026

Frontend Web Development using React

Project Submission Template

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DATE: 24/06/2026

TITLE: Mini E-Commerce System

GITHUB LINK : <https://github.com/kavi5670583-NM/Kavitha-T>

Mini E-Commerce System



ABSTRACT

The Mini E-Commerce System is a web-based application designed to simulate the core functionalities of an online shopping platform. The system allows users to register, browse products, add items to the cart, place orders, and make payments in a simplified and secure manner. It also provides an admin panel for managing products, categories, orders, and users. The project is developed using modern web technologies to ensure a user-friendly interface, efficient performance, and responsive design. This system helps in understanding e-commerce workflows, database management, and basic transaction processes, making it a useful learning tool for students and beginners in web development.



PROBLEM STATEMENT

Traditional shopping methods often require customers to visit physical stores, which can be time-consuming and inconvenient. Many small businesses and beginners lack a simple and affordable platform to manage products and process orders efficiently. Existing e-commerce platforms may be complex and difficult for students to understand and implement.

The Mini E-Commerce System aims to address these issues by providing a simple and user-friendly web application that enables users to browse products, add items to the cart, place orders, and manage basic transactions.

The system also includes an admin module for managing products and orders.

This project helps in understanding the fundamental concepts of e-commerce, database management, and web application development while providing a practical solution for online shopping operations.

OBJECTIVES

The main objective of this project is to develop a web application that provides a simple, efficient, and user-friendly solution to manage and process data effectively. The system is designed to automate manual tasks, reduce errors, and improve overall productivity. It aims to deliver accurate results, ensure data security, and provide a seamless experience for users. Additionally, the project focuses on creating a scalable and maintainable application that can be easily updated and extended to meet future requirements.

The project focuses on developing a web application that addresses the specific requirements identified. It includes features such as user registration and authentication, data management, and reporting functionalities. The system is designed for ease of use, efficiency, and reliability. The project covers requirement analysis, system design, development, testing, and deployment. It aims to provide a secure and scalable solution that meets the needs of the intended users while ensuring maintainability and future enhancement possibilities.

TECHNOLOGIES USED



Frontend

- React.js
- CSS3
- JavaScript



Backend

- Node.js
- Express.js



Database

- MongoDB



Tools

- VS Code
- GitHub
- NPM



Others

- REST API
- JSON
- Bootstrap

APPLICATION FLOW

1. User Registration

User creates an account or logs in to the system.

2. Browse Products

User browses products by category, search, or filters.

3. Add to Cart

User selects products and adds them to the cart.

4. View Cart

User reviews selected items, quantity, and price.

5. Checkout

User enters shipping details and selects payment method.

6. Payment

User completes payment through chosen method.

7. Order Confirmation

System confirms order and generates order ID.

8. Order Processing

Admin processes the order and prepares for shipping.

9. Shipping

Order is shipped and tracking information is updated.

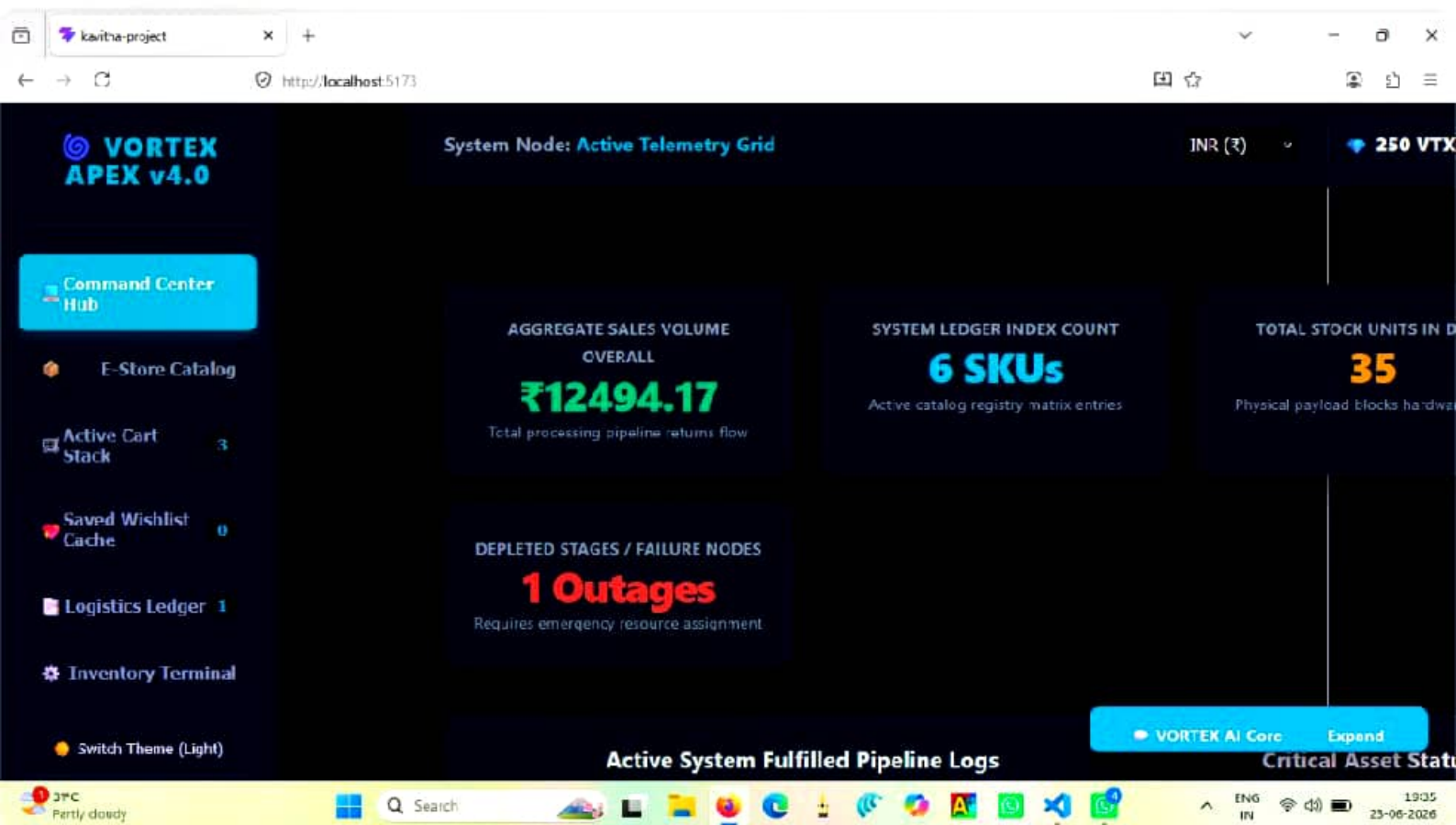
10. Order Delivered

User receives the order and can review the product.

KEY: REACT CONCEPT USED

1. **Components** – Building reusable UI parts.
2. **JSX** – Writing HTML inside JavaScript.
3. **Props** – Passing data from parent to child components.
4. **State** – Managing dynamic data in a component.
5. **useState** – Hook used to manage state in functional components.
6. **useEffect** – Handling side effects like API calls and subscriptions.
7. **Event Handling** – Responding to user interactions.
8. **Conditional Rendering** – Rendering UI based on conditions.
9. **Lists & Keys** – Rendering lists of items efficiently with unique keys.
10. **Forms** – Handling user input and form submission.
11. **Controlled Components** – Managing form data using state.
12. **React Router** – Navigating between pages in the application.
13. **Custom Hooks** – Reusing logic across components.
14. **Context API** – Sharing global data without prop drilling.
15. **Fragments** – Grouping multiple elements without extra nodes.
16. **Memoization (React.memo)** – Optimizing performance by preventing unnecessary re-renders.
17. **useContext** – Accessing context data in components.
18. **Error Boundaries** – Handling UI errors gracefully.
19. **Code Splitting** – Splitting code to improve performance.
20. **React DevTools** – Debugging and inspecting React components.

Expected output:



levithe-project x +

← → ↻ http://localhost:5173

VORTEX APEX v4.0

System Node: Active Telemetry Grid

INR (₹) 250 V

Command Center Hub

E-Store Catalog

Active Cart Stack 3

Saved Wishlist Cache 0

Logistics Ledger 1

Inventory Terminal

Switch Theme (Light)

Query data tags or product tags...

All Categories ▾

All Categories

Electronics

Accessories

Apparel

Fitness

Standard Structure

Vortex Pro ANC Audio Matrix

₹12494.17 Add Node

Apex Chronograph Steel Horizon

₹9163.00 Add Node

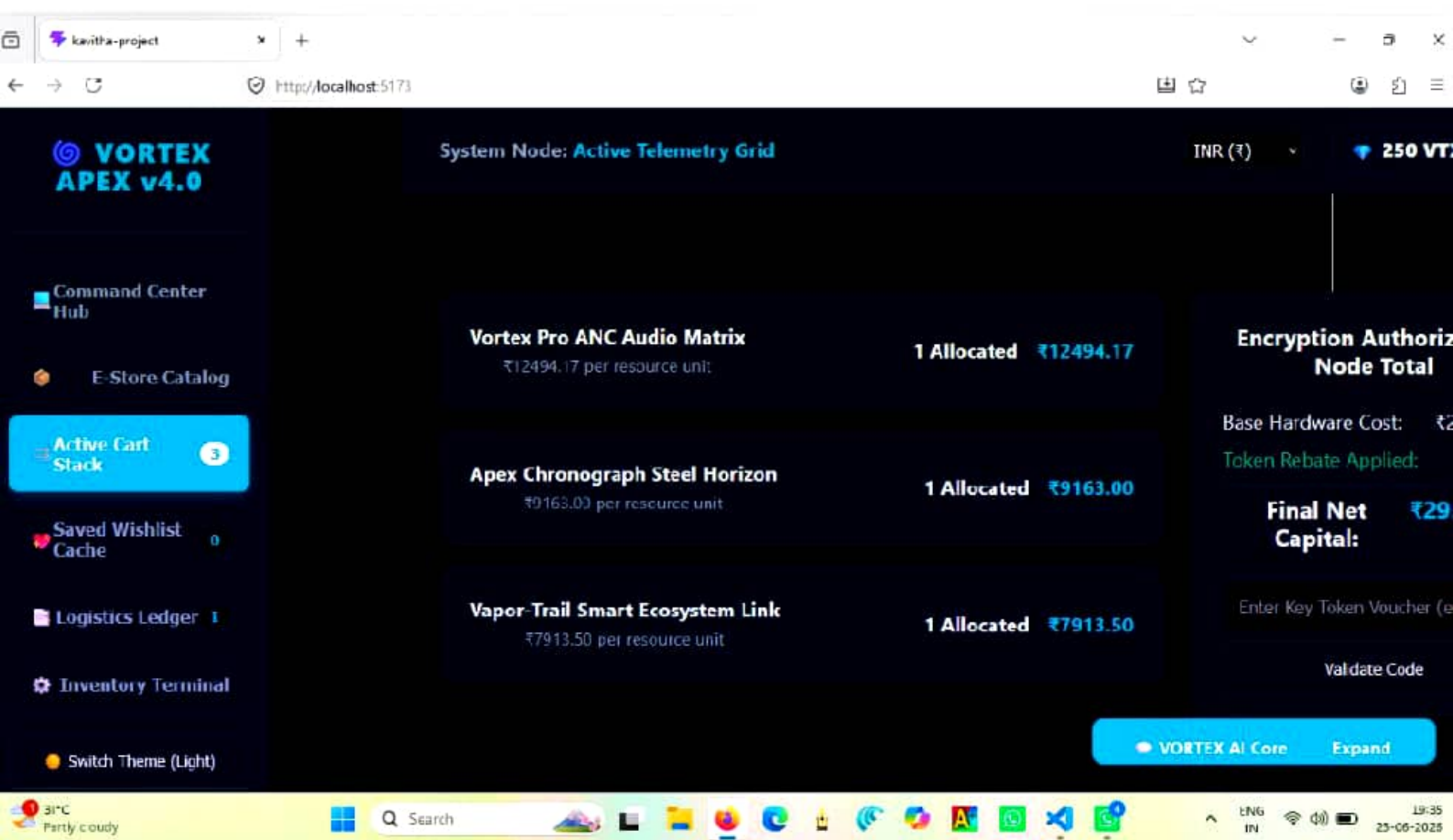
Quantum V3 Mechan

VORTEX AI Core Expand

31°C Partly cloudy

Search

ENG IN 19:3 23-06-202



CHALLENGES FACED

During the development of this project, several challenges were encountered. Understanding complex concepts and implementing them in a practical way was initially difficult. Integrating different modules and ensuring smooth communication between them required significant time and debugging. Handling data efficiently while maintaining accuracy and consistency was another major challenge. Additionally, designing a user-friendly interface that meets user requirements and works across different devices required extra effort. Time management and meeting deadlines while ensuring quality also posed a challenge throughout the project. However, these challenges helped improve problem-solving skills and provided valuable learning experiences.

Through this project, I gained a deeper understanding of the concepts and technologies used in developing a web application. I learned how to analyze requirements, design user-friendly interfaces, and implement the system effectively. This project helped improve my problem-solving skills, coding abilities, and knowledge of tools and technologies used. I also gained experience in testing, debugging, and ensuring the accuracy and reliability of the system. Overall, this project has enhanced my technical skills, boosted my confidence, and prepared me to work on real-world applications in the future.

CONCLUSION

In conclusion, this project was a valuable learning experience that helped in understanding the practical application of concepts learned. The system developed meets the intended objectives and provides an efficient solution to the problem identified. Through this project, technical skills were improved, and knowledge in planning, designing, and implementing a system was gained. Overall, the project has been a successful effort that has enhanced both technical and analytical abilities, which will be beneficial for future endeavors.